SPEAR3 LLRF Upgrade

Subsystem Integration and Migration Plan

Document ID	Doc M
Revision	R0
Status	Draft
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Classification	Tier 1 — Project integration and migration plan
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Date	3 November 2026

This plan analyzes how each subsystem of the SPEAR3 LLRF Upgrade can be put into service inside the running legacy RF station, which subsystems are independent of the others and which are not, and how to proceed if some subsystems are deferred. It is based on the project status and decisions of 31 October to 3 November 2026, the upgrade design (Doc 0), the legacy documentation, the original drawings and design notes, and the legacy EPICS source code. Statements that still need confirmation are tagged **[O*n*]** and collected in the Open Items register.

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Revision History

Revision	Date	Scope of Change	Summary
R0	2026-11-03	Draft for review	Initial issue: legacy coupling analysis, subsystem independence assessment, software plan, staged migration plan and deferral scenarios, based on the project status and decisions of 31 October to 3 November 2026.

Open Items — Still To Be Confirmed

The statements below come from drawings, design notes and source code and have not yet been confirmed on the installed hardware or with the responsible engineer. Each is tagged **[O*n*]** where the plan relies on it; the page column links to that point.

ID	Item	What is needed	Needed before	Page
O1	RF MPS program	Confirm that the ControlLogix program reproduces the PLC-5 behavior, including the hardwired outputs to the RF Local Display Panel and the heartbeat, and confirm the status of both ladder programs.	RF MPS cutover	p. 17
O2	Legacy IOC changes	Confirm that the legacy database can be edited and reloaded, whether the SNL programs can still be rebuilt (EPICS 3.14.11, vxWorks), and whether a test IOC can be set up.	Any legacy IOC change, including the minimal coordinator	p. 12
O3	RF Local Display Panel logic	Recover the equations of the programmable logic (GAL22V10) that forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort, by reading the device or by a truth-table test of the panel.	Interface Chassis taking over any of these outputs	p. 11
O4	Beam-abort endpoint	Trace where the panel's rear-panel BEAM ABORT connector is cabled, and agree with the SPEAR MPS group on the signal level, polarity, latching and the action it causes.	Interface Chassis production design; VXI retirement	p. 16
O5	Legacy heartbeat	Measure the period and duty cycle of the legacy AB heartbeat and confirm that they suit the Interface Chassis watchdog.	Interface Chassis in series	p. 16
O6	Slot 5 permit loop	Trace the SPEAR MPS and orbit-interlock wiring into VXI Slot 5 and check the source current budget for a second (Interface Chassis) input.	Interface Chassis in series	p. 16
O7	ARC4 trip point	Define the ARC4 output interface (Doc 0 O5) and choose the interim permit point in the legacy chain from the Fast Interlock Chassis and AIM drawings.	Enabling the ARC4 trip	p. 14
O8	2021 LLRF9 setup	Recover how the drive was switched, how interlocks were handled and how turn-on was sequenced in the 2021 parallel test.	LLRF9 drive stage	p. 23

ID	Item	What is needed	Needed before	Page
O9	RIO adapter emulation	Confirm with Rockwell whether a ControlLogix module can present the PLC-5's DCM tables to the legacy RIO scanner.	Choice of the RF MPS EPICS path	p. 17
O10	HVPS unit transfer	Write the list of connections transferred when the active HVPS unit is switched, for one legacy and one upgraded unit.	HVPS stage	p. 18

1 Purpose, Scope and Basis

1.1 Purpose

The SPEAR3 LLRF Upgrade replaces the control electronics of the SPEAR3 RF station and keeps its high-power plant [R1]. By the end of October 2026 nearly all new hardware is built, the EPICS-side software has not been written [R38], and the station must keep delivering beam between scheduled downs. This plan answers three questions:

1. Which upgrade subsystems can be put into service inside the running legacy station on their own, which need only small, well-defined changes to the legacy system, and which depend on other subsystems?
2. For each subsystem: where does it connect to the legacy station, what must change in the legacy system, and how is the change reversed if it fails?
3. In which order should the subsystems go into service, and how should the project proceed if some of them are deferred for lack of people, funds or installation time?

1.2 Scope

In scope are the upgrade subsystems of Doc 0 [R1]: the AFT ARC4 arc detection system, the Interface Chassis and Waveform Buffer, the RF MPS PLC, the HVPS controller with its firing boards, regulator and PPS Interface Box, the klystron heater supply, the Galil tuner motion controller, the two LLRF9 units and the control software. Also in scope are the parts of the legacy station they replace or connect to. Out of scope are the high-power plant that is kept (klystron, HVPS power section, waveguide, cavities) and the proposed solid-state amplifier project [R26], except where it affects priorities (§12).

1.3 Basis

- Upgrade design: Doc 0 R4 [R1], with Doc P [R4] and Doc T [R5] for the loop and tuner analyses.
- Legacy system: Doc L [R2] and Doc I [R3]; J. Sebek's design notes [R6]–[R9], [R27], [R28], [R39], [R42]; the original drawings and spreadsheets [R10], [R11], [R25], [R29]–[R34]; the legacy EPICS source code [R14]–[R23], [R36], [R41]; the LLRF9 vendor documentation [R24], [R40].
- Project status and decisions reported by the project lead from 31 October to 3 November 2026 [R38] (§3); subsystem leads from the April 2026 project schedule [R13].

Reading of the software status. "No software has been written yet" [R38] is read here as the EPICS side: IOC databases, operator screens, sequencing and the coordinator. The PLC ladder programs are taken as recorded earlier: the RF MPS program written and tested on SPEAR3 without RF power [R1], and the HVPS program about 95 % complete in April 2026 [R13]. If either ladder program is not final, the corresponding stage (§10.2) moves later.

1.4 Terms

Legacy station The RF station as it operates today: VXI LLRF crate, PLC-5 RF MPS, SLC-500 HVPS controllers, 1746-HSTP1 tuner controllers, Fast Interlock Chassis 340-308, AIM, RF Local Display Panel 340-310 and the variac heater supply.

RF Local Display Panel The legacy panel 340-310 in B132 (rack B132-12) whose schematic, SD-340-311-01, is titled "PEP-II RF System Local Panel" [R30]; it is also called the Local Control Chassis. It links the Fast Interlock Chassis, the RF MPS PLC-5 and the HVPS controller, displays the station status, and contains the hardware logic that forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort (§4.3).

Seam A point where a legacy element and its replacement present the same interface to the rest of the station, so that one can be exchanged for the other without changing anything else.

Monitor-only A new subsystem is installed and observed, with no output in any control or trip path.

Series insertion A new element is placed in a permit path so that it can remove a permit but never create one.

Cutover, fallback A cutover puts a subsystem into the operating configuration during a scheduled down; the fallback is the tested procedure that restores the previous configuration.

Hard, soft dependency A hard dependency must be met before a cutover; a soft dependency makes the cutover easier or better but is not required.

2 Summary

2.1 Main Conclusions

1. **Four couplings hold the legacy station together**: one Remote I/O daisy chain, one hardwired protection chain, one RF drive source (the RFP) and one supervisory IOC with a single trip wire (§4). A new subsystem can join the running station on its own only where its interface to these couplings stays the same, at a seam (§5).
2. **The RF Local Display Panel is an active protection element.** Its logic forms the SCR ENABLE and KLYSTRON CROWBAR fibers to the HVPS controller and the beam abort to the SPEAR MPS (§4.3). Its equations are not documented and must be recovered before the Interface Chassis takes over these outputs.
3. **Most subsystems have a seam.** The ARC4, the heater supply and the LLRF9 tuner control through the Galil can go in with little or no change to the legacy system. The RF MPS and the HVPS controller can replace their legacy counterparts behind unchanged field wiring and unchanged PV names; the HVPS controller goes in together with the PPS Interface Box. The Interface Chassis can be added in series with the legacy protection chain; it shares one chassis with the Waveform Buffer, and the two go in together.
4. **As the drive source, the LLRF9 is the most dependent subsystem.** Every fast RF-off path of the legacy station ends on a VXI backplane line that only the RFP obeys: arcs and reflected power through the AIM, and the SPEAR MPS beam permit and orbit interlock through VXI Slot 5. With the LLRF9 driving the klystron, a SPEAR MPS or orbit trip would no longer remove RF in hardware. The LLRF9 as drive source therefore needs the Interface Chassis (or an interim permit adapter) and a minimal coordinator; it does not need the new RF MPS, HVPS controller or heater (§6.9.2).
5. **The LLRF9 is useful long before it drives the klystron.** It can take over the tuner loops, driving the Galil directly, while the RFP still drives the klystron (§6.8). This works in normal operation, at turn-on, at turn-off and after trips, and it needs no change to the legacy state machine: the legacy tuner loops are switched off and their alarms adjusted. A separate EPICS control of the Galil is not needed.
6. **Software is now the critical path.** Nearly all hardware is built, but no EPICS-side software exists. If

the legacy PV names are kept and the legacy SNL programs stay in charge, each early cutover needs only its subsystem IOC and a small change to the legacy database. The coordinator, which is the IOC B132-IOCRF itself, is needed in a minimal release for the LLRF9 drive stage and in full for VXI retirement (§9).

7. **Retiring the VXI crate is the only step that needs everything at once.** It removes the RIO scanner, the AIM and Slot 5, the RFP and the IQAs, the VXI CPU and the station's 476 MHz reference distribution, and with them the Fast Interlock Chassis and the RF Local Display Panel; the IOC B132-IOCRF moves to new hardware (§7.3).
8. **Some legacy functions have no assigned home yet**: control of the klystron focus and bucking-coil supplies, the filament timer bypass, the HV turn-on gate, the no-restart latch after hard arcs and others. The beam abort to the SPEAR MPS is required and becomes an Interface Chassis output (§8).
9. **Two changes touch the legacy protection and need configuration control**: the RF hookup of the LLRF9 and the Waveform Buffer, because every tap lowers a legacy signal and moves its calibration and trip thresholds, and the drive-amplifier patch that selects the RFP or the LLRF9 (§6.9.1).
10. **Recommended strategy**: use the running legacy station as the integration test bed, in a series of small cutovers. Each cutover sits behind a seam, is preceded by monitor-only operation where possible and has a tested fallback (§10). A single integration at Test Stand 18 is not an option while the test stand is not in normal operation [R26].

2.2 Independence at a Glance

The subsystems fall into three classes. Class A subsystems are nearly independent. Class B subsystems replace a legacy element behind a seam and need work on the legacy database, the RIO chain or the PPS wiring. Class C subsystems depend on other subsystems.

Table 3: Independence of the upgrade subsystems.

Subsystem	Class	Joins the legacy station by	Hard prerequisites	Fallback
ARC4 arc detection	A	Running monitor-only, then tripping at one legacy permit point	Fiber test, false-trip record, trip point	Disable the trip
Heater supply	A	Replacing the variac stage under the legacy enable and monitors	60 Hz sync circuit, heater IOC, a long down	Variac chassis
Tuner (LLRF9 driving the Galil)	A	LLRF9 tuner loops drive the Galil while the RFP still drives the klystron	LLRF9 hookup and cross-calibration; legacy tuner loops off	Motor cables back to the SS2000 drivers
RF MPS	B	Replacing the PLC-5 behind the same field wiring and PV names	Legacy-faithful program, IOC, legacy database change, RIO splice, trip test	PLC-5
HVPS controller and PPS Interface Box	B	Replacing the SLC-500 on the off-duty HVPS unit and taking the PPS chains out of the PLC, in one change	ADSO approval and final design of the PPS box, PPS certification, IOC, legacy database change; firing board and regulator unless decoupled	Other HVPS unit

Subsystem	Class	Joins the legacy station by	Hard prerequisites	Fallback
Interface Chassis and Waveform Buffer (one chassis)	B/C	Installed together: the Interface Chassis as a series element in the legacy protection chain (B), later the central permit hub (C); the Waveform Buffer recording beside the legacy diagnostics	Production unit, IOC, features for running beside the legacy chain, beam-abort output; RF taps and HVPS signal cables	Bypass jumper
LLRF9 as drive source	C	Replacing the RFP as the drive source	Interface Chassis, minimal coordinator, interlocked RF transfer switch	RFP as hot standby
VXI retirement	C	—	All stages, full coordinator on the new hardware of B132-IOCRF, homes for orphan functions	None practical

3 Project Status Baseline

Table 4: Status of the upgrade subsystems, 31 October and 2 November 2026.

Subsystem	Hardware	EPICS-side software	Source
LLRF9 (two units)	Installed in B132; RF cabling to be completed in the coming weeks. One unit will run the four tuner loops and drive the Galil directly.	Vendor IOC, upgraded to EPICS 7 in June 2026	[R38], [R1]
Galil DMC-4143	Chassis built; tested on the SPEAR3 tuner motors under an EPICS control written by M. Dunning. That control is not carried forward, because the LLRF9 drives the Galil directly.	None needed beyond the LLRF9	[R38], [R37]
ARC4	Assembled; parts from the same vendor as the existing arc-detection hardware; accepts the existing tunnel fibers; ready to install	None	[R38], [R1]
Interface Chassis and Waveform Buffer	One chassis with two boards; prototype ready for review, then production	Waveform Buffer data acquisition and screens working (April schedule); none for the Interface Chassis	[R38], [R13]
RF MPS (ControlLogix)	Assembled; online for software development; EPICS communication working	None	[R38], [R1]
HVPS controller (CompactLogix)	Online for software development; EPICS communication working; FCOG1200 firing boards ordered; regulator board being redesigned	None	[R38], [R1]
Heater supply (Genesys AC)	Running on a test stand with a load; external 60 Hz sync circuit in design	None	[R38], [R1]
PPS Interface Box	Design of the initial version done; documentation mostly done; ADSO approval and the final design still needed	Not applicable	[R38]

Subsystem	Hardware	EPICS-side software	Source
Control software	TID software review completed	No EPICS-side software written	[R38]

The status moves the critical path from hardware to software, reviews and approvals, and installation windows. The items not covered by the status update (firing boards and regulator, heater sync circuit) are taken as recorded in Doc 0 [R1].

4 How the Legacy Station Is Coupled

4.1 Overview

The legacy station is held together by four couplings (Figure 1):

1. one Allen-Bradley Remote I/O (RIO) daisy chain, through which every PLC datum reaches EPICS (§4.2);
2. one hardwired protection chain—AIM, Fast Interlock Chassis, RF Local Display Panel and the HVPS fibers—and the permit inputs of VXI Slot 5 (§4.3);
3. one RF drive source, the RFP, with the three IQA modules as the measurement front end (§4.4);
4. one supervisory IOC whose SNL programs and alarm tree coordinate the station (§4.5).

A new subsystem can join the running station on its own only where its interface to these four couplings stays the same.

4.2 The Remote I/O Chain

The VXI slot-1 Allen-Bradley 6008-SV1R scanner, mounted through an HP E1407A adapter, is the master of a single RIO link [R2]. The scanner configuration defines three racks [R15], and the legacy database binds them as listed in Table 5 [R36].

Table 5: Racks on the legacy RIO link.

Rack	Size	Device and adapter	Location
1	Full	HVPS SLC-500 through a 1747-DCM (last node)	B118
2	3/4	Tuner stepper chassis 340-315 with four 1746-HSTP1 modules (middle node)	B132
3	1/4	RF MPS PLC-5 through a 1771-DCM (first node)	B132

Physically the link is one daisy chain [R9]. The scanner cable goes to the DCM in the main PLC-5 crate. A daisy-chain cable from the DCM terminals goes to the tuner chassis. From the tuner chassis terminals a cable goes to a terminal box above the HVPS termination tank, and from there a long cable runs to the HVPS SLC-500 in B118. The main PLC-5 crate reaches a second PLC-5 crate below it over its own link.

Consequences for the migration:

- Every PLC datum and command reaches EPICS only through this chain and the VXI IOC.
- RIO is a multidrop link wired as a daisy chain. A powered-down node loses only its own data, but disconnecting a node's terminals cuts every node downstream. Replacing the PLC-5 (first node) or the tuner chassis (middle node) therefore means splicing the chain through; replacing the HVPS SLC-500 (last node) means moving the terminating resistor.

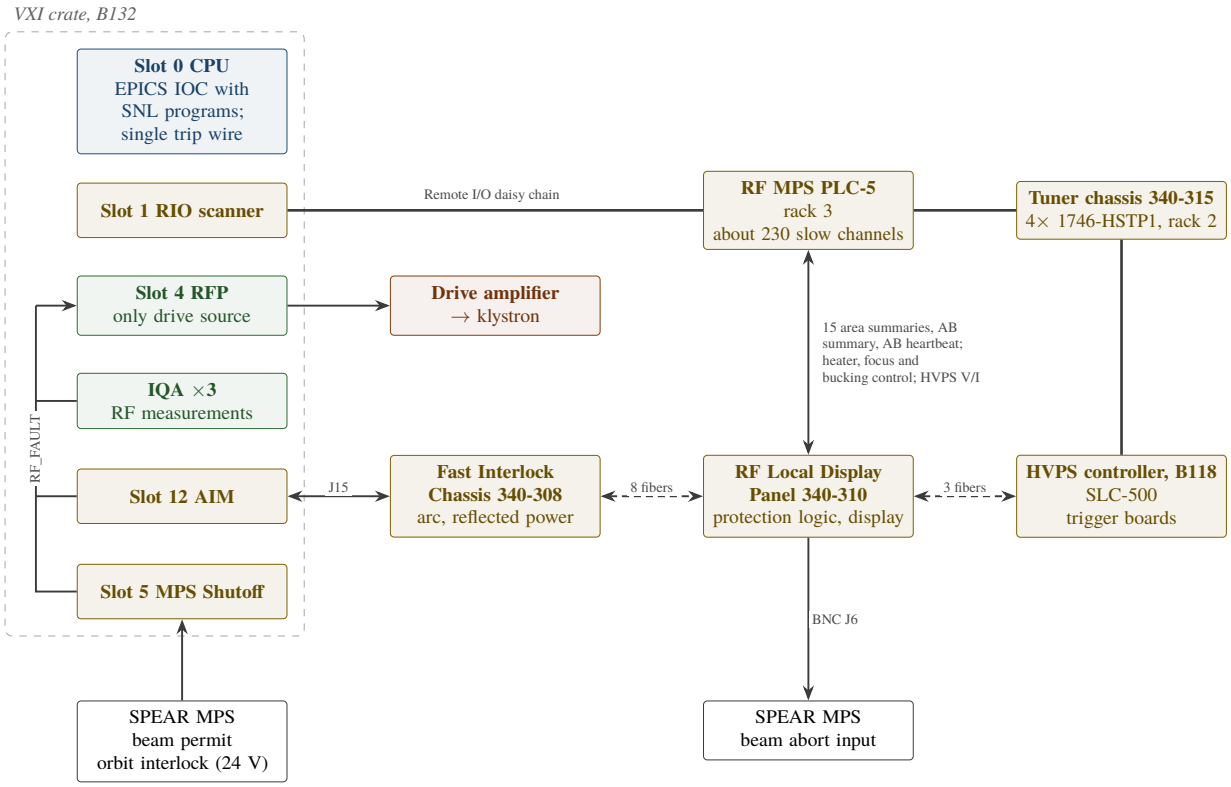


Figure 1: Couplings of the legacy station that a new subsystem must respect. The Remote I/O link is one daisy chain from the VXI scanner through the RF MPS, tuner and HVPS racks. Every fast RF-off path ends on the RFP through the open-collector backplane line RF_FAULT. The AIM, the Fast Interlock Chassis, the RF Local Display Panel and the HVPS controller form one hardwired protection chain. The three HVPS fibers (SCR ENABLE, KLYSTRON CROWBAR, STATUS) run between the panel in B132 and the HVPS controller in B118, and the panel's logic forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort to the SPEAR MPS. Dashed arrows are fibers.

- If a rack stops answering, its records go to INVALID severity and the alarm summaries carry that to the station trip wire. A rack that is retired must be removed from `config.ab` and its records re-pointed in the same change (§9.4).
- Retiring the VXI crate removes the scanner. By then every node must be on Ethernet or bridged.

4.3 The Hardwired Protection Chain

The protection chain is documented at the level of connectors and fibers in J. Sebek's notes [R9], [R39] and in the original drawings [R30], [R33], [R34]:

- **AIM and Fast Interlock Chassis.** Connector J15 of the chassis is cabled to J3 of the AIM (VXI slot 12). Of its 25 signals, 23 are driven by the AIM; the two inputs are the AB summary fault and the AB heartbeat [R9], [R32].
- **Fast Interlock Chassis and RF Local Display Panel.** Eight fibers [R33]. Six leave the Fast Interlock Chassis and carry the AIM's commands: FILAMENT_ON (station online request), SOLENOID_ON (filament and focus request), HVPS_ON (HVPS on request), BEAM_ABORT (VXI beam abort), FILAMENT_TIMEOUT (filament timeout bypass) and FAULT_RESET. Two return the PLC-5's AB

SUMMARY and AB HEARTBEAT [R9].

- **RF Local Display Panel and HVPS controller.** Three fibers run between the panel in B132 and the trigger interconnect boards of the HVPS controller in B118 [R9], [R33], [R39]. SCR ENABLE (the panel's "HVPS triggers enable", lit to enable) goes to the Right Side Trigger Interconnect board (SD-730-793-07); the thyristor triggers fire only while it is lit. KLYSTRON CROWBAR (lit for no crowbar) goes to the Left Side board (SD-730-793-08); when it goes dark the crowbar fires. STATUS comes from the Left Side board and is lit while the controller's "24 V fault/enable" output is high and no crowbar is commanded; the panel receives it as the "HVPS beam abort" [R7], [R30].
- **The panel's logic.** A programmable logic device (GAL22V10) forms SCR ENABLE, KLYSTRON CROWBAR and the beam abort from the HVPS on request, the VXI and HVPS beam aborts, the main and bucking solenoid faults, the waveguide pressure and collector summaries from the PLC-5 and the station online key switch; the beam abort and the crowbar are latched until reset [R30], [R39]. The beam abort leaves on a rear-panel BNC connector (J6) for the SPEAR MPS [R30]. The equations of the logic are not documented. **[O3]**
- **RF Local Display Panel and RF MPS PLC-5.** Two 25-pin connectors, J2 and J3, wired to cross-connect block X530 in rack B132-14 (Appendix C) [R9], [R10], [R34]. They carry the AIM's requests to the PLC-5 as 24 V lines, the PLC-5's 15 area summaries and its AB summary and heartbeat, filament, focus and bucking-coil supply control and interlocks, and the buffered HVPS voltage and current that the PLC-5 uses to compute collector power. The panel also displays the station status and carries the station online/offline and local/remote key switches [R30].
- **VXI Slot 5 ("MPS Shutoff").** The SPEAR MPS beam permit and the orbit interlock (24 V each); dropping either asserts RF_FAULT [R2], [R3]. Which module is installed in Slot 5 is still open in Doc I (O1).
- **RF_FAULT.** An open-collector VXI backplane line. The AIM, the IQAs, Slot 5, the CLK module and the RFP can assert it; the RFP then removes the drive [R2].

Beam abort. The state machine forces a beam abort through the AIM whenever it turns the station off, enters TUNE or switches the direct loop, and resets it when the station reaches ON or PARK with no faults [R21]. The panel adds the HVPS STATUS and the solenoid faults in hardware. The signal therefore tells the SPEAR MPS that the RF station is not ready for beam.

> Every fast RF-off path of the legacy station ends on the RFP. High voltage is removed in hardware through the RF Local Display Panel's SCR ENABLE and KLYSTRON CROWBAR fibers or inside the HVPS controller. A SPEAR MPS or orbit trip reaches the HVPS only through the RF cut and the state machine, unless the PLC-5's collector-power protection acts first: after any trip the state machine spends about 1 s in TUNE and 5 s ramping the HVPS down before it removes the triggers and the RF switch [R21].

4.4 The RF Drive and Measurement Path

- The RFP is the only drive source. The patch-panel record of the 2021 LLRF9 configuration at B132-11 lists the RFP output (J12) and the drive-amplifier input (J13) as separate patch cables next to the LLRF9 reference (J11) [R11]: the drive source is selected by a patch cable.
- The three IQA modules measure the 24 RF signals. The legacy loops and alarms use their data: drive power for the HVPS loop, probe and forward phases for the tuner loops, and reflected and forward powers for the alarms.
- In 2021, 20 dB couplers were inserted upstream of the legacy inputs for the cavity forward and reflected signals and the klystron reflected signal (insertion loss 0.38–0.43 dB); power splitters were added on

the four probes and the klystron forward signal (0.28–0.46 dB). The legacy calibration constants were changed to compensate [R11], [R12].

- In that test one LLRF9 received the four probe signals, the four cavity forward signals and the klystron forward signal, and its reference was taken from output RF3 of the VXI RF distribution (CLK) module [R11].

4.5 The Supervisory IOC and Its Alarm Tree

- The VXI slot-0 processor runs EPICS under vxWorks as the IOC B132-IOCRF. At boot it loads one expanded database, `db/srf1.db`, restores saved settings, configures the RIO scanner from `/cmd/config.ab` and starts nine SNL program instances: the state machine, four tuner loops, the HVPS loop, the DAC loop, a calibration sequence and the message logger [R14]. In the upgrade this IOC remains the station IOC and is the coordinator; it moves to new hardware when the VXI crate is retired [R38].
- The state machine trips on one summary, `SRF1:STNOFF:SUMY:STAT`. It collects the VXI latch, the HVPS off summary, the online key switch of the RF Local Display Panel, the AB summary latch and the PLC-5's own station summary bit; turn-on is blocked by a similar summary, `SRF1:STNON:SUMY:STAT` [R21], [R23]. Cavity, tuner and temperature alarms reach only the station status display.
- A database change is a text edit and a reboot. An SNL change needs the EPICS 3.14.11 vxWorks cross-development environment, whose availability is not confirmed. **[O2]**
- The bindings that matter for the migration are listed in Appendix B. Three stand out. The HVPS loop reads the RFP's direct-loop state and module severity [R20]. The tuner loop uses five fields of the stepper-motor records and sets its target through the position-control record [R19], [R22]. The state machine homes the tuners by processing `SRF1:CAVTUNR:LOOPON:RESET` and then waits a fixed 60 clock ticks without checking completion [R21].

4.6 Data the RF MPS PLC Provides to Other Subsystems

The PLC-5 is the station's slow data system as well as an equipment-protection PLC. About 230 template instances in the legacy database take their data from its DCM tables (Appendix B): about 110 temperatures, about 40 interlock bits, about 30 analog channels and about 50 summary, module and digital status points [R17], [R18]. Several belong to other subsystems:

- the tuner potentiometer positions (`SRF1:CAVnTUNR:POSN`, $n = 1$–4) and the tuner temperatures;
- the stepper drivers' bulk supply voltage (`SRF1:CAVTUNR:VOLT`, low-low alarm at 20 V), whose latch feeds the tuner summary [R17], [R23];
- filament voltage and current with their PLC-5 trip limits, and main focus and bucking-coil supply voltage and current;
- klystron and circulator ion-pump currents and window temperatures;
- the HVPS voltage and current as seen by the PLC (the collector-power inputs) and the HVPS oil temperature [R16].

A change to the RF MPS must keep all of these, and a change to the tuner, heater or HVPS must check how these channels react.

5 Integration Seams and Rules

5.1 Seams

Table 6 lists the seams found in the legacy station. Each allows one or more subsystems to be exchanged without changing the rest of the station.

Table 6: Integration seams.

Seam	What crosses it today	In the upgrade	What it allows
HVPS fiber trio	SCR ENABLE, KLYSTRON CROWBAR and STATUS between the RF Local Display Panel (B132) and the HVPS controller (B118)	The same three fibers and meanings [R6], [R7]	Exchanging the HVPS controller and the B132 interlock electronics independently, in either order
Beam-abort output	BNC from the RF Local Display Panel to the SPEAR MPS	Interface Chassis output	Moving the beam abort without changing the SPEAR MPS side
Data source of EPICS records	Records in `srf1.db` bound to RIO racks	Channel Access links to new IOCs	Exchanging the RF MPS and HVPS controllers without SNL changes
Legacy PV names	Names used by the SNL programs, alarm tree, screens, archiver and MATLAB tools	Kept in the IOC B132-IOCRF until the full coordinator; the HVPS names for good [R38]	Any replacement that keeps the names, units and alarm limits
RF MPS field-wiring arms	1771 field I/O	Reused by the Rockwell conversion kit [R1]	Exchanging the RF MPS hardware with a quick way back
Slot 5 permit loops	SPEAR MPS and orbit-interlock permits (24 V)	Moved to the Interface Chassis	Series insertion of the Interface Chassis or the ARC4
RF couplers	Probe, forward and reflected taps upstream of the IQAs	Extended to the LLRF9 and Waveform Buffer inputs	Running the LLRF9 and Waveform Buffer beside the IQAs
Drive-amplifier input	RFP output	Interlocked RF transfer switch	A reversible LLRF9 drive cutover
Tuner motor cables	SS2000 drivers	Galil, driven by the LLRF9	The Galil with a way back
Heater J1 path	Filament enable and voltage and current monitors	Kept until the RF MPS takes heater control	The Genesys AC under legacy control
PPS connector, K4 and Ross coil drive	PPS permits and readbacks	PPS Interface Box	The PPS box together with the HVPS controller, independent of the Interface Chassis and the LLRF9

5.2 Rules for Every Cutover

1. **Legacy names are the interface.** A replacement keeps the PV names, units, scaling and alarm limits of the records it replaces until the full coordinator takes over; the HVPS keeps its names for good.
2. **Series insertion only.** A new element in a permit path can remove a permit but never create one. If an element regenerates a signal instead of interrupting it, a failure-mode analysis comes first.
3. **Monitor-only first.** Where the subsystem allows it, it runs monitor-only for a run period before any output is connected.
4. **One change per window.** A scheduled down changes one subsystem, so that any new behavior has one cause. The Interface Chassis with the Waveform Buffer, and the HVPS controller with the PPS Interface

Box, each count as one change.
5. **Tested fallback.** Every cutover has a written fallback procedure, tested before the window, and the legacy hardware stays in place and intact until the next stage has run without it.
6. **Configuration control.** Every hybrid configuration is recorded: connections, calibration constants, database version and the state of every jumper and bypass.
7. **People.** Two stages that need the same person are not scheduled in the same window (§10.4).

6 Subsystem Assessments

Each assessment uses the same headings. *Joining the legacy station* gives the recommended first service mode, *Fallback* the tested way back and *Verdict* the class of Table 3.

6.1 Arc Detection (AFT ARC4)

Status. The ARC4 chassis is assembled and ready to install. Its parts come from the same vendor as the existing arc-detection hardware, and it accepts the existing tunnel fibers [R38]. Twelve detector modules are procured: ten cover the four cavity windows (two fibers each), the klystron output window and the circulator; two are spares. Two system modules and a test module form the external interface [R1].

Legacy counterpart. Arc inputs on the Fast Interlock Chassis with per-channel latching in the AIM. This circuitry is largely disconnected [R9] and the optical arc protection is not in use [R1]; today arcs are caught only through the reflected-power comparators. The ARC4 therefore adds protection rather than replacing a working function.

Joining the legacy station.
1. Install monitor-only: fibers plugged in, permit output in no trip path, status and fired-channel identity read into EPICS through a small networked digital I/O module or the Interface Chassis IOC (SW4).
2. At the next tunnel access, light-test every fiber end to end (none has been used in service) and exercise the electronics with the test module.
3. Record the ARC4 trips with beam over a run period before any trip is enabled.
4. Enable the trip at one point, in order of preference: (a) the Interface Chassis production unit (§6.3); (b) an interim hardwired point in the legacy chain, either a series contact in the AB summary line into the RF Local Display Panel, which reaches the Fast Interlock Chassis and the AIM, or a series contact in the Slot 5 permit loop, which removes RF within microseconds but appears as a SPEAR MPS trip in the legacy alarms; (c) a PLC input as a last resort, with a scan time of 10–20 ms and a program change. The choice between the interim points needs the Fast Interlock Chassis and AIM drawings [R31], [R32]. **[O7]**

Prerequisites. Hard: the ARC4 output interface (Doc 0 O5), the fiber test, a recorded monitor-only period. Soft: the Interface Chassis production unit.

Pitfalls. Nuisance trips, which may be why the original optical system was disconnected; confusing diagnostics if the trip enters through Slot 5; the latch reset must be assigned (ARC4 front panel, IOC or Interface Chassis).

Fallback. Remove the series contact or disable the trip; the ARC4 stays as a monitor.

Verdict. Class A. The ARC4 can join the legacy station almost independently; the only design decision is where its trip enters.

6.2 Waveform Buffer

Status. Board complete; data acquisition and screens reported working in the April 2026 schedule [R13]; shares a chassis with the Interface Chassis, whose prototype is ready for review, and goes into service with it [R38].

Legacy counterpart. The history buffers of the AIM, IQAs and RFP with the SNL fault files, and the collector-power calculation in the PLC-5.

Joining the legacy station. Installed with the Interface Chassis in stage S4 (§10.2.5); the Waveform Buffer itself is monitor-only, beside the legacy diagnostics. Its six RF channels (circulator load reflected, station reference, waveguide loads 1 to 3 reflected and, proposed, the klystron drive power, which is on neither LLRF9 unit [R1]) need taps (§6.9.1). Its HVPS voltage and current are available in B132 at the buffers of the RF Local Display Panel (Appendix C) and must be taken through high-impedance buffers. The two inductor voltages need new cables from the HVPS. Before its comparator trip is used, its collector-power calculation is compared with the PLC-5's over a run period.

Prerequisites. The Interface Chassis production unit; RF taps, HVPS signal cables, IOC (SW7).

Pitfalls. Loading the panel's buffers that feed the PLC-5's collector-power calculation. In the shared chassis, a Waveform Buffer fault must reach the Interface Chassis only through the defined comparator outputs.

Fallback. Disconnect its inputs, with its comparator outputs to the Interface Chassis disabled.

Verdict. Class A in function, because it is monitor-only; its timing follows the Interface Chassis production unit, with which it is installed.

6.3 Interface Chassis

Status. Prototype ready for review; production unit to follow [R38]. It is one 3U chassis with the Waveform Buffer [R1], and the two are installed and integrated together [R38].

Target design. The central hardware permit hub [R6], [R1]. Inputs: LLRF9 status, the HVPS STATUS fiber, the RF MPS summary permit, heartbeat and reset, the Waveform Buffer comparators, the SPEAR MPS and orbit-interlock permits (24 V), the ARC4 and four expansion ports. Outputs: LLRF9 enable, the SCR ENABLE fiber, the KLYSTRON CROWBAR fiber (kept lit), and input and output states and a first-fault record for the RF MPS. It has its own IOC and absorbs the functions of the Fast Interlock Chassis and the AIM [R1], and with them the protection logic of the RF Local Display Panel.

Beam-abort output. The project requires the beam abort to the SPEAR MPS to continue [R38]. In the legacy station it leaves the RF Local Display Panel on a rear-panel BNC, formed from the VXI beam abort (forced and reset by the state machine through the AIM), the HVPS STATUS fiber and the main and bucking solenoid faults, and latched until reset [R21], [R30]. The Interface Chassis takes this output over with the same inputs, the coordinator's force and reset replacing the AIM's, latched, with a fail-safe convention agreed with the SPEAR MPS group and with its state reported to the RF MPS. J. Sebek's specification provides no output to the SPEAR MPS [R6] and must add it. Where the legacy output is cabled and what the SPEAR MPS does with it is not shown on any RF drawing. **[O4]**

Joining the legacy station as a series element.

- Inputs without a source yet: LLRF9 status (jumpered until the LLRF9 drives); RF MPS summary and heartbeat, taken from the PLC-5's AB summary and AB heartbeat lines at the RF Local Display Panel (J3-09 and J3-21); **[O5]** reset from the legacy interlock reset request (panel J2-23), the front panel or the IOC.
- SPEAR MPS and orbit-interlock permits: paralleled from the Slot 5 wiring after a check of the source current budget, or routed through the Interface Chassis, which returns an isolated contact to Slot 5. **[O6]**
- Outputs the legacy station can use: (i) an isolated contact in series in the Slot 5 permit loop, which is simple and can only remove the permit; (ii) a regenerated SCR ENABLE equal to the panel's SCR ENABLE AND the Interface Chassis permits, which needs a receiver for the panel's fiber, a STATUS pass-through to the panel and a failure-mode analysis.
- J. Sebek notes that the cross-connect diagram numbers connector J3 in reverse order to the panel schematic [R39]; any connection at J3 is checked on the hardware first.

Production review checklist.

1. Features for running beside the legacy chain: STATUS fiber pass-through or repeater to the RF Local Display Panel; receiver for the panel's SCR ENABLE fiber; one isolated dry-contact permit output; a bypass or simulate jumper per input; a configurable heartbeat window; an external reset input.
2. The beam-abort output to the SPEAR MPS described above.
3. Logic equivalence: the Interface Chassis reproduces the panel's SCR ENABLE, KLYSTRON CROWBAR and beam-abort logic, and every deliberate difference is documented and reviewed (O3).
4. Bench tests that close Doc 0 O1 (LLRF9 logic levels) and O5 (ARC4 output interface) with the real units, which are now on hand.
5. A decision to keep the RF Local Display Panel through the transition (recommended) or to absorb its I/O now (Appendix C).
6. A hardware output for the HV turn-on gate (§8).
7. The five-step LLRF9 and HVPS recovery sequence [R6] and its interaction with the legacy reset and automatic restart.

Prerequisites. As a series element: the production unit, its IOC (SW7), the checks O5 and O6, and the Waveform Buffer's RF taps and HVPS signal cables. As the hub: the LLRF9 as drive source, the RF MPS in upgrade mode, the ARC4 interface, the beam-abort endpoint (O4) and the panel logic (O3).

Fallback. Bypass jumper; remove the series contact.

Verdict. Class B as a series element; Class C as the hub.

6.4 RF MPS (ControlLogix)

Status. Assembled; program written and tested on SPEAR3 without RF power [R1]; online for software development with EPICS communication [R38].

Legacy counterpart. The PLC-5 with 1771 I/O on RIO rack 3, the first node of the chain. It is the station's slow data system (§4.6) and drives hardwired outputs to the RF Local Display Panel: 15 area summaries, two of which (collector and waveguide pressure) also feed the panel logic, the AB summary and heartbeat, filament, focus and bucking-coil control and the filament timer bypass (Appendix C) [R10], [R39].

Joining the legacy station.

- The program must first reproduce the PLC-5 behavior: summary polarity, heartbeat timing, collector-power calculation and limits, and heater, focus and bucking-coil control. The functions that exist only in the upgrade (Interface Chassis summary, heartbeat and reset, Interface Chassis status inputs, heater analog ramp, Waveform Buffer inputs) belong in a second, switchable mode. **[O1]**
- The program keeps a legacy data image (§9.5).
- EPICS path: the legacy records are re-pointed to a new EPICS 7 IOC (recommended, §9.4); RIO adapter emulation is an alternative if it exists. **[O9]**
- RIO chain: spliced through to racks 2 and 1; rack 3 removed from `config.ab`.
- Verification in the cutover window: the full trip-matrix test (Doc L O3) and every channel of §4.6 present with the same scaling and alarm limits.

Prerequisites. Legacy-faithful program, SW1, SW5, trip-test procedure.

Fallback. The PLC-5 stays intact: field-wiring arms moved back, previous `config.ab` and `srf1.db` restored.

Verdict. Class B. Independent of the Interface Chassis, the LLRF9 and the HVPS controller; on the critical path for VXI retirement.

6.5 HVPS Controller

Status. CompactLogix online with EPICS communication [R38]; ladder program about 95 % complete in April 2026, with klystron cathode control remaining [R13]; FCOG1200 firing boards ordered; regulator board being redesigned [R1].

Legacy counterpart. One SLC-500 controller per HVPS unit (SPEAR1 and SPEAR2). One unit is energized at a time, and the roles are exchanged in scheduled downs, typically in April and August [R1]. The controller is RIO rack 1, a full rack and the last node of the chain [R15], [R36], and it is in the PPS chain through rungs 0016 and 0017 [R2], [R3]. Its slot-5 output 0 (24 V fault/enable) enables the thyristor triggers on the command bus of the trigger interconnect boards in the same enclosure [R7].

Joining the legacy station.

- The fiber trio to the RF Local Display Panel is unchanged if the trigger interconnect and driver boards are kept or reproduced exactly. STATUS is lit when the controller's 24 V fault/enable output is high and no crowbar is commanded. A dark SCR ENABLE inhibits the triggers of phases A and C, while B^+ and B^- keep firing so that the filter inductors can discharge [R7], [R8].

- The EPICS footprint is small: eleven rack-1 records in the HVPS database plus 32 status bits (Appendix B) [R16]. A new IOC with a legacy data image serves them; the legacy records are re-pointed, rack 1 is removed from `config.ab` and the chain is re-terminated.
- Two HVPS units: while one unit is legacy and the other upgraded, the legacy IOC must serve either (two database variants or a selector), and the switch-over procedure needs a list of the connections transferred. The procedure is a stub today [R35]. **[O10]**
- The HVPS voltage and current monitor outputs to the RF Local Display Panel and the PLC-5 keep their scaling, because the collector power is computed from them.
- The local control panel (touch panel, PERMIT key switch, EMERGENCY OFF button with two contacts) is reproduced; the EMERGENCY OFF test of Doc L (O12) is part of the commissioning.
- The new PLC does not carry the PPS chains: the PPS Interface Box is installed and certified with the new controller, in the same down (§6.6).
- Option to decouple the controller from the firing board and regulator: install the CompactLogix first with the existing FCOG6100 board set and regulator card, reproducing their analog setpoints, and exchange the firing board and regulator in a later down, if spares and the analog interface allow.
- PV names: the HVPS keeps its current PV names in the IOC B132-IOCRF, and all HVPS control stays in EPICS, with no Python layer [R38] (§9.6).

Prerequisites. PPS box approved, installed with the controller and certified; SW1 and SW6; FCOG1200 and regulator, or the decoupled option. Test Stand 18 would help but is not in normal operation [R26].

Fallback. Switch to the other HVPS unit.

Verdict. Class B, together with the PPS Interface Box. Independent of the LLRF9, the Interface Chassis and the RF MPS; gated by the PPS approval.

6.6 PPS Interface Box

Status. The design of the initial version is done and the documentation is mostly done; ADSO approval and the final design are still needed [R38]. The design (B. Morris) is a Bud enclosure with four relays, status LEDs and one PPS-lockable connector, with two permit and two readback channels; it is the standard design used for the 6575 modulator and gallery systems [R1].

Legacy counterpart. PPS 1 enters an SLC-500 input and the supply of the output that drives contactor K4; PPS 2 enters another input, and rung 0016 drives the Ross grounding-switch coil from a PLC output. Chain 2 has no hardware protection against a PLC output stuck on [R3], [R29].

Joining the legacy station. The box goes into service together with the new HVPS controller, in the same down and under one PPS certification (§10.2.6); it is not installed with the legacy controller. The box drives K4 and the Ross coil directly, as designed [R1]. One behavior changes: in the legacy ladder the Ross coil needs PPS 1 and PPS 2 (rung 0016) [R2], while the box drives it from PPS 2 alone [R1]; this belongs in the PPS review. Two HVPS units need one box each, or a cable that is moved at the switch-over, which already breaks the PPS chain [R35].

Prerequisites. ADSO approval, final design and documentation, Radiation Safety Work Control Form, PPS certification tests.

Fallback. As for the HVPS controller: the other HVPS unit, with its legacy PPS wiring.

Verdict. Class B, as one change with the HVPS controller, which it gates; independent of every other subsystem. It takes the PLC out of the PPS chain.

6.7 Klystron Heater Supply (TDK-Lambda Genesys AC)

Status. Running on a test stand with a load [R38]; external 60 Hz sync circuit in design [R1].

Legacy counterpart. A motor-driven variac, a 10:1 step-down isolation transformer, a solid-state relay and meters in the filament control chassis [R2], [R25]. The filament request from the AIM reaches the chassis through the Fast Interlock Chassis, the RF Local Display Panel and the PLC-5, which also commands the variac position [R2], [R9], [R10]. The voltage and current monitors return to PLC-5 analog inputs, and a filament timer can be bypassed from the AIM through the FILAMENT_TIMEOUT fiber [R9].

Operating point and limits. The klystron runs at 68 V and 7.3 A [R1]. The PLC-5 trips the filament outside 62.4–76.2 V and 5.24–9.77 A; these limits are held in the PLC-5 and read back into EPICS (`SRF1:FILAMENT:VOLT:ULIM` and `SRF1:FILAMENT:VOLT:LLIM`, and the same for the current) [R17]. The live EPICS alarm limits are given in Table 7 [R38].

Table 7: Live filament limits [R38].

Channel	PLC-5 low trip	LOLO	LOW	HIGH	HIHI	PLC-5 high trip
Voltage (V)	62.4	63.8	66.5	72.1	76.2	76.2
Current (A)	5.24	6.72	7.01	7.59	7.88	9.77

The archived substitution file holds different alarm limits (95–128 V, 4.7–6.5 A) [R17]: the live limits come from the IOC's saved settings, so any database generated for the new systems takes its limits from the live configuration (§9.2).

Target design. The RF MPS drives the supply's analog programming input, Ethernet carries monitoring and enable, and the legacy meters are kept [R1].

Joining the legacy station before the RF MPS upgrade.

- The legacy filament-on signal drives the supply's remote enable.
- The warm-up ramp comes from the supply's own soft start or from the heater IOC (SW3) and follows the legacy warm-up sequence.
- The legacy voltage and current monitors stay, so the PLC-5 heater interlock and its limits are unchanged; the variac position command is unused.
- The cleanest mechanical form replaces only the variac, motor and relay stage and keeps the chassis transformer, meters and connector J1.

Test-stand acceptance (closes Doc 0 O6): command set and IOC; regulation and waveform at 68 V and 7.3 A into the real transformer type, well inside the limits of Table 7; lock to the 60 Hz sync; enable and inhibit wiring; monitor-output scaling against the PLC-5 inputs; output held through a mains dip and through loss of the network link.

Pitfalls. The cathode needs a long down to cool and re-warm.

Prerequisites. Sync circuit, SW3, a long down.

Fallback. The variac chassis, kept in place.

Verdict. Class A.

6.8 Tuner Control (LLRF9 Tuner Loops Driving the Galil DMC-4143)

Status. The Galil chassis is built and was tested on the SPEAR3 tuner motors under an EPICS control written by M. Dunning [R37], [R38]. The project has decided that the LLRF9 tuner loops drive the Galil directly through the LLRF9's built-in Galil driver [R38]. The separate EPICS control is therefore not carried forward, which saves its development and maintenance.

Legacy counterpart. Four 1746-HSTP1 modules in chassis 340-315 on RIO rack 2, SS2000 drivers, and four SNL tuner loops using IQA phases [R2], [R5], [R41]. The loop runs every 2 s in every station state except OFF; it moves a tuner in TUNE and ON_CW while the klystron forward power is above a minimum and holds in ON_FM. At turn-on the state machine homes the tuners to their ON positions [R21], [R28].

Configuration.

- The master LLRF9 unit carries the four probe, the four cavity forward and the klystron forward signals and runs all four tuner loops, as in the vendor's four-cavity configuration and in the 2021 test [R11], [R38], [R40]. Only this unit commands the Galil.
- The LLRF9 tuner loop is a PID controller on the phase between probe and forward signals, computed every 0.1 s, with a deadband, a loop sign, a load-angle offset and a minimum forward power below which it does not move the tuner [R24], [R40]. Through its Galil driver it commands the tuner velocity and reads the encoder position, the limit switches and an analog input, to which the tuner potentiometer can be wired [R24].
- The driver's screen is labeled for the DMC-21X3; it is bench-tested with the installed DMC-4143 before the stage (§10.2.2).

Operation with the RFP still driving. Table 8 compares the legacy tuner control with the LLRF9 driving the Galil in each operating condition.

Table 8: Tuner control in each operating condition.

Condition	Legacy tuner control	LLRF9 driving the Galil	Needed
Normal operation (ON or TUNE)	SNL loop every 2 s on the IQA load angle, offset zero [R5], [R41]	PID every 0.1 s on the LLRF9's own probe and forward phases [R40]	Legacy loops off; LLRF9 offsets calibrated against the legacy setpoint in shadow
Turn-on	The state machine homes the tuners to their ON positions and waits 60 ticks before enabling HV [R21]	No preset move: the tuners start where they stopped, and the loop starts once the forward power exceeds its minimum	Homing request becomes a no-op; turn-on from a full-current detuning verified
Commanded turn-off	Station OFF; the loop holds [R41]	Forward power below the minimum; the loop holds	None

Condition	Legacy tuner control	LLRF9 driving the Galil	Needed
Trip	As turn-off, after the RF cut	As turn-off; the few samples taken while the fields decay are limited by the deadband and the gains	Trip tests with beam confirm that the tuners do not move during the decay
LLRF9 IOC or unit down	—	The tuners hold; the detuning then follows beam current and temperature	Restore the IOC, use a spare unit [R42] or fall back to the legacy cables

Turn-on from a detuned position. After a trip at full current a tuner stays near the beam-loaded detuning of about -36 kHz, roughly one half-bandwidth of the loaded cavity [R4]. With the legacy homing gone, the next turn-on starts there, with more reflected power than today until the loop has pulled the cavity in. At 20 to 30 kHz/mm this is 1.2 to 1.8 mm of travel, which takes well under a second at the legacy tuner speed [R4], [R5]. Commissioning checks that the reflected power stays within its limits during the turn-on ramp; if not, the minimum forward power is lowered so the loop starts earlier.

Changes on the EPICS side. The legacy state machine needs no change. In the legacy database:

- The four legacy tuner loops are switched off (`SRF1:CAVTUNR:LOOP:CTRL`). That record and the loop status in its LOOP_OFF state are MAJOR alarms that roll up into the station severity summary, not into the trip summaries; their severities or the alarm configuration are adjusted [R22], [R23].
- The stepper records and the 1746-HSTP1 chassis stay powered and connected: they keep rack 2 alive on the RIO chain and are the fallback. The state machine's homing request then moves only the disconnected legacy drivers, which is harmless.
- The tuner potentiometers, temperatures and bulk-supply channel stay on the PLC-5; their alarms reach the station status display only [R23].
- The LLRF9 tuner loop status, limit switches and positions are added to the alarm handler and the archiver (SW8).

Pitfalls.

- Without a separate EPICS control there is no path to the Galil when the LLRF9 IOC is down, other than the Galil's own command interface.
- The Galil's software limits must lie inside the legacy drive limits, from -28 mm to 18 mm, and the mechanical travel restriction of ± 10 mm recorded in Doc L [R2], [R22].
- The LLRF9 phase offsets must reproduce the legacy detuning setpoint; they are calibrated in the shadow comparison.

Prerequisites. LLRF9 RF hookup and cross-calibration (SW8); LLRF9 Galil driver configured and bench-tested (SW2).

Fallback. Move the motor cables back to the SS2000 drivers and switch the legacy tuner loops on again.

Verdict. Class A. Independent of the RF MPS, the HVPS controller, the heater and the Interface Chassis; depends on the LLRF9 only, and needs no change to the legacy state machine.

6.9 LLRF9 Field Control

Status. Two units installed in B132; RF cabling to be completed in the coming weeks [R38]; vendor IOC upgraded to EPICS 7 in June 2026 [R1]. In 2021 an LLRF9 ran in parallel with the legacy system [R12]. Four units were bought, so two are spares [R42]. The master unit takes the four probe, four cavity forward and klystron forward signals, and its outputs OUT0 and OUT2 will drive the klystron; the slave takes the four cavity reflected, klystron reflected, circulator-load forward and three waveguide-load forward signals [R1], [R38].

The LLRF9 has three roles, which can be commissioned in this order: measurement and diagnostics, tuner control (§6.8) and drive source.

6.9.1 RF Hookup as a Change to the Legacy System

- Every tap or splitter lowers a legacy signal: the IQA calibrations and the trip thresholds expressed in IQA units move. Reflected-power and other protection thresholds are rechecked after any change.
- The coupler and splitter hardware from 2021 [R11] is reused where possible, and the patch-panel record is updated as part of the change. The slave's circulator-load and waveguide-load forward signals had no tap in 2021 [R11] and need new ones.
- The LLRF9 needs its own reference distribution before the VXI crate is retired, because the reference is taken from the VXI CLK module today.
- The drive-amplifier input becomes an interlocked RF transfer switch, not a patch cable, with its position read into EPICS.

6.9.2 The LLRF9 as Drive Source

Table 9 shows which legacy protection paths still act when the LLRF9 drives the klystron.

Table 9: Legacy protection with the LLRF9 as drive source.

Fault source	Legacy fast action	With the LLRF9 driving
Arc or reflected power at the Fast Interlock Chassis	RF_FAULT through the AIM; HV removed through the RF Local Display Panel fibers as far as its logic acts (O3)	HV removal unchanged; RF_FAULT no longer reaches the drive
IQA amplitude comparators	RF_FAULT	No action; the LLRF9's own input interlocks take this role
SPEAR MPS beam permit, orbit interlock	Slot 5 asserts RF_FAULT; the state machine removes HV later	No hardware action; HV removed only by the state machine
RF MPS trip	AB summary to the AIM; collector and waveguide summaries into the panel logic	HV removal as in the legacy station; no hardware RF cut
HVPS controller trip	HV removed; STATUS goes dark and raises the panel's beam abort; drive stays on until the state machine turns it off	Unchanged

Consequences.

- The Interface Chassis, at least as a series element with the LLRF9 enable wired, is a hard prerequisite. An interim permit relay box that opens the LLRF9 interlock input from Slot 5's permits and the legacy summaries is possible as an emergency measure only.
- A minimal coordinator is a hard prerequisite (§9.7). The legacy HVPS loop uses the RFP's direct-loop state and module severity [R20]; with the RFP idle it would regulate on the wrong variable or alarm. Turn-on, the HVPS loop on LLRF9 drive power and the fault response must therefore change in the IOC B132-IOCRF, which is the coordinator; while it still runs on the VXI CPU, changes to its SNL programs depend on O2.
- The RFP is kept as a hot standby on the transfer switch, with the IQAs still measuring, so the legacy diagnostics remain available.
- The 2021 test recorded no drive switching, interlock or turn-on procedure, and its configuration needs to be recovered. **[O8]**
- Performance acceptance: in 2021 the ripple rejection at ± 360 Hz and ± 720 Hz was 16–21 dB for the LLRF9 against 24–27 dB for the legacy system, and the synchrotron line was at $9\,657$ Hz with a quality factor of 16.7 against $8\,932$ Hz and 11.1 [R12]. The acceptance criteria for these quantities come from the loop design of Doc P [R4].

Verdict. Class C. Needs the Interface Chassis and the minimal coordinator; does not need the new RF MPS, HVPS controller or heater.

6.10 Control Software

Status. TID review completed; no EPICS-side software written [R38].

Assessment. Software is not one subsystem but one work package per subsystem plus the coordinator. Each subsystem IOC is small and mostly standard; the tuner needs no IOC of its own because the LLRF9 drives the Galil. The legacy IOC changes are small but need an owner and a test IOC. The coordinator is the largest item and is needed only from the LLRF9 drive stage on. The controls group leads every work package, with technical support from the subsystem leads. The plan is in §9.

7 Dependency Analysis

7.1 Dependency Matrix

Table 10 lists, for each cutover, what it needs from the other parts of the project. H is a hard prerequisite, S a soft prerequisite.

Table 10: Dependency matrix (H hard, S soft).

Cutover	IC	MPS	HVPS	LLRF9	Coord.	Legacy IOC	Window
ARC4, monitor-only	—	—	—	—	—	—	Access
ARC4, trip enabled	S	—	—	—	—	—	Down
Heater (S3)	—	S	—	—	—	—	Long down
Tuner (S1)	—	—	—	H[a]	—	S	Down
RF MPS (S2)	—	n/a	—	—	—	H	Down, trip test

Cutover	IC	MPS	HVPS	LLRF9	Coord.	Legacy IOC	Window
HVPS controller and PPS box (S5)	—	—	n/a	—	—	H	HVPS unit exchange; PPS certification
Interface Chassis and Waveform Buffer (S4)	n/a	S	—	—	—	—	Down
LLRF9 drives (S6)	H	—	—	n/a	H, SW9	H	Down and machine time
VXI retirement (S7)	H	H	H[b]	H	H, SW10	n/a	Long down

IC: Interface Chassis in series. MPS, HVPS: the new RF MPS and the new HVPS controller. Coord.: coordinator release (SW9 minimal, SW10 full). Legacy IOC: change to the IOC B132-IOCRF. [a] The LLRF9 as measurement system and tuner controller only. [b] Or an RIO bridge to the legacy SLC-500.

7.2 Independence Classes

The matrix confirms the classes of Table 3. The ARC4 monitor-only installation and the heater depend on nothing else. The tuner depends only on the LLRF9, plus alarm adjustments in the legacy database. The RF MPS, the HVPS controller with the PPS Interface Box, and the Interface Chassis with the Waveform Buffer depend on legacy IOC work or on their own reviews and approvals, but not on each other. Only the LLRF9 as drive source and the VXI retirement depend on other subsystems.

7.3 What Retiring the VXI Crate Requires

Retiring the VXI crate removes at once the RIO scanner, the AIM, Slot 5, the RFP, the IQAs, the CLK module and the CPU on which the IOC B132-IOCRF runs, and with the AIM's commands gone the Fast Interlock Chassis and the RF Local Display Panel have no purpose. It therefore needs:

- the RF MPS and the tuner off the RIO link (S2, S1) and the HVPS controller replaced (S5) or bridged by an RIO scanner module in the RF MPS chassis;
- the LLRF9 as drive source with the Interface Chassis as the hub (S6);
- new hardware for the IOC B132-IOCRF, running the full coordinator (SW10);
- a home for every orphan function (§8), including a separate $476\,\mathrm{MHz}$ reference distribution;
- the protection logic, the beam abort and the display of the RF Local Display Panel taken over by the Interface Chassis, and its PLC-5 connections by the RF MPS.

8 Legacy Functions Without an Assigned Home

The legacy station performs the functions in Table 11, which the upgrade design does not yet assign to any subsystem. Each needs a home, or a decision that it is no longer needed, before its legacy implementation is removed.

Table 11: Legacy functions without an assigned home.

#	Function	Legacy implementation	Proposed home	Needed before
1	Beam abort to the SPEAR MPS (required [R38])	AIM force and reset by the state machine; RF Local Display Panel logic adding the HVPS STATUS and the solenoid faults; rear-panel BNC [R21], [R30]	Interface Chassis output; force and reset from the coordinator (§6.3)	S7
2	Main focus and bucking-coil supply control and interlocks	PLC-5 through the RF Local Display Panel [R10]	RF MPS outputs	S7
3	Filament request and timer bypass	AIM, Fast Interlock Chassis, RF Local Display Panel and PLC-5 [R9]	RF MPS and heater IOC	S7
4	HV detected, RF detected, local/remote, online/offline	RF Local Display Panel to PLC-5 [R10]	RF MPS inputs	S7
5	Buffered HVPS voltage and current for the collector-power calculation	RF Local Display Panel buffers [R10]	HVPS controller to RF MPS	S5 or S7
6	HV turn-on gate	AIM “HVPS on” request into the RF Local Display Panel logic	Coordinator with an Interface Chassis permit	S6
7	No-restart latch after hard arcs	SNL forced-fault latch [R3]	Coordinator	S6
8	Reference power, crate and RIO health monitors	SNL and VXI records	Reference distribution monitor; IOC health records	S7
9	Fault capture and MATLAB fault tools	AIM, IQA and RFP history buffers; MATLAB tools	LLRF9 and Waveform Buffer post-mortem; ported tools	S6
10	Readbacks for other subsystems	PLC-5 channels (§4.6)	RF MPS, with the same names	S2
11	476 MHz reference distribution	VXI CLK module	Separate reference distribution chassis	S7

9 Software Plan

9.1 Why Software Is Now the Critical Path

The hardware of every subsystem except the HVPS firing board, the regulator, the PPS box and the heater sync circuit is built or in hand, and no EPICS-side software exists [R38]. Each cutover of §10.2 needs at least its subsystem IOC or configuration, and the LLRF9 drive stage and the VXI retirement need the coordinator. Software therefore sets the pace of the migration, and its plan follows the stage order.

9.2 Principles

1. **Minimum software per cutover.** The legacy SNL programs stay in charge until the LLRF9 drive stage, so each early cutover needs only its subsystem IOC and a legacy database change.
2. **Generate rather than write.** The legacy substitution files hold names, units, alarm limits and data addresses for every legacy channel. With a legacy data image in the new PLCs (§9.5) they generate most of the RF MPS and HVPS databases. Alarm limits and other settings are taken from the IOC's saved settings, because the archived files differ from the live values (§6.7).
3. **Standard modules and vendor software first.** The LLRF9's own IOC, including its tuner loops and Galil driver; a standard protocol driver for the Genesys AC; the EtherNet/IP driver for the ControlLogix and CompactLogix.
4. **Two coordinator releases**, each with one adapter per subsystem that can talk to either the legacy or the new hardware, so that any deferral order is supported (§9.7).

9.3 Work Packages

Every work package is led by the controls group, with technical support from the lead of the subsystem it serves [R38]. Table 12 names that lead, as in the April 2026 schedule [R13].

Table 12: Software work packages, led by the controls group.

WP	Content	Needed for	Relative size	Technical support
SW0	PV naming policy, legacy compatibility rules, one interface control document per seam	All stages	Small	Project lead
SW1	Owner for the legacy IOC; test IOC; procedure to edit and reload `srf1.db`; check of the SNL build	S1, S2, S5, S6	Small to medium	J. Sebek
SW2	LLRF9 tuner loops driving the Galil: driver configuration, limits and bench test; legacy tuner loops off and their alarms adjusted	S1	Small	M. Dunning
SW3	Heater IOC (Genesys AC over Ethernet)	S3	Small	B. Morris
SW4	ARC4 status readout (read-only)	S0	Small	R. Lion, K. Vedyanina
SW5	RF MPS IOC generated from the legacy substitution files, the saved settings and the legacy data image; legacy records re-pointed	S2	Medium	M. Larrus
SW6	HVPS IOC behind the current HVPS names in B132-IOCRF; selector in the legacy database for the legacy or upgraded HVPS unit	S5	Small	M. Larrus
SW7	Interface Chassis and Waveform Buffer IOC (status, first fault, beam abort, waveforms)	S4	Medium	J. Sebek, D. Zhang
SW8	LLRF9 configuration, cross-calibration, screens, alarms and archiving	S0, S1, S6	Medium	J. Sebek, M. Dunning
SW9	Minimal coordinator in B132-IOCRF on the VXI CPU (§9.7)	S6	Medium to large	J. Sebek
SW10	Full coordinator in B132-IOCRF on its new hardware, operator screens, post-mortem tools, ported MATLAB tools	S7	Large	J. Sebek

9.4 Adapting the Legacy IOC

When a legacy PLC is replaced, the legacy records that read it must take their data from the new system. Table 13 compares three methods.

Table 13: Methods for connecting the legacy IOC to a replaced PLC.

Method	How	For	Against
RIO adapter emulation	A ControlLogix module presents the PLC-5's DCM tables to the legacy scanner	No legacy IOC change	Feasibility unconfirmed; keeps the RIO dependency
Re-pointed legacy records (recommended)	The legacy records keep their names, alarm limits and summaries; their data source changes from RIO to Channel Access links to the new IOC	Alarm tree and screens unchanged; the new IOC keeps neutral names	Legacy database must be edited and tested
New IOC serving the legacy names	The records are removed from the legacy database and recreated, with the same names and limits, in the new IOC	The legacy IOC loses the records cleanly	Every alarm limit and summary link must be reproduced exactly

Each legacy database change follows the same procedure: edit; test on the test IOC; apply in a down with the previous `srf1.db` and `config.ab` kept; verify the alarm tree and the SNL behavior; record the configuration. A named owner for the legacy IOC and a test IOC are prerequisites for stages S2, S5 and S6, and help with S1.

9.5 Legacy Data Image in the New PLCs

The PLC-5 and the SLC-500 exchange their data with EPICS through the RIO images of their DCMs, and the legacy records address these images by link, adapter, card and signal [R16], [R36]. If each new PLC program keeps tag arrays with exactly this layout, maintained by the ladder logic, then:

- the legacy records can be re-pointed one for one;
- the new IOC databases can be generated from the legacy substitution files;
- old and new values can be compared word by word during commissioning.

The cost is a small amount of ladder logic, and the image is retired with the VXI crate.

9.6 PV Naming Policy

Two IOCs serving one name give undefined Channel Access behavior. Proposed policy:

- while a legacy record exists, its name belongs to the IOC B132-IOCRF;
- the HVPS keeps its current names, such as `SRF1:HVPS:VOLT:CTRL` and `SRF1:HVPS:VOLT:RBCK` [R16], [R38];
- other new IOCs use prefixes that cannot collide with legacy names for their native records, such as `SRF1:MPS:` for the RF MPS and `SRF1:WFBUF:` for the Waveform Buffer in Doc 0 [R1];
- exceptions are deliberate and recorded, and legacy names that are not kept are retired with the VXI crate.

9.7 Coordinator Scope

The coordinator is the IOC B132-IOCRF [R38]. The minimal release runs there while the IOC is still on the VXI CPU, so a change to its SNL programs needs the legacy build environment (O2); the full release runs on its new hardware. All HVPS functions stay in EPICS, with no Python layer [R38].

Minimal release (SW9), for S6.

- LLRF9 turn-on and turn-off sequence: ramp, loop close, transition from low power.
- HVPS supervisory loop on LLRF9 drive power: the legacy HVPS loop [R20], changed to regulate on the LLRF9 drive power and keeping its states, status codes and tolerance checks.
- Fault response: HVPS ramp-down and SCR disable, beam-abort force and reset through the Interface Chassis (replacing the AIM's), the no-restart latch after hard arcs, Interface Chassis reset.
- Adapters for the subsystems that are still legacy at that point.

Full release (SW10), for S7. On the new hardware of B132-IOCRF: station state machine, automatic recovery, station-level calibration, probe balancing (an LLRF9 function [R24]), fault manager and post-mortem from the LLRF9 and Waveform Buffer data, operator and expert screens, alarm and archiver configuration, and ported MATLAB tools.

10 Migration Plan

10.1 Strategy

The station must keep running, and Test Stand 18 is not in normal operation [R26]. The plan therefore uses the running legacy station as the integration test bed: a series of small cutovers, each behind a seam, each preceded by monitor-only operation where possible, each in a scheduled down and each with a tested fallback. The rules of §5.2 apply to every stage.

10.2 Stages

10.2.1 S0: Preparation and Monitor-Only Installation

- **When:** now; no effect on beam.
- **Work:** decisions D1–D11 (§13); SW0 and SW1 and the subsystem work packages SW2–SW8 started; RF hookup as a controlled change (§6.9.1); ARC4 installed monitor-only; heater acceptance on the test stand; production review of the Interface Chassis and Waveform Buffer; PPS box documentation completed and submitted for ADSO approval; 2021 LLRF9 setup recovered; RF Local Display Panel logic recovered (O3) and beam-abort endpoint traced (O4).
- **Exit:** LLRF9 cross-calibrated against the IQAs; ARC4 data in EPICS; decisions recorded.

10.2.2 S1: Tuner — LLRF9 Driving the Galil

- **Entry:** RF hookup done; the tuner part of SW8; SW2 with the LLRF9 Galil driver tested with the DMC-4143 on the bench; loop parameters (gains, deadband, minimum forward power, offsets) set from a shadow comparison of the LLRF9 tuner phase with the legacy load angle.
- **Work:** motor cables moved to the Galil; legacy tuner loops switched off and their alarms adjusted; LLRF9 loops closed one cavity at a time, first at low power.
- **Verification:** tuner positions against the potentiometers; detuning and reflected power against legacy values at several beam currents; turn-on from a full-current detuning; a trip with beam.
- **Fallback:** motor cables back to the SS2000 drivers; legacy tuner loops on.

10.2.3 S2: RF MPS — ControlLogix with Legacy PVs

- **Entry:** legacy-faithful program; SW1 and SW5; legacy database change tested; RIO splice prepared; trip-matrix procedure.

- **Work:** field-wiring arms moved; RIO chain spliced; rack 3 removed from `config.ab`; legacy records re-pointed.
- **Verification:** full trip matrix; every PLC-5 channel present with correct scaling and limits; panel summaries, heartbeat and collector and waveguide summaries accepted by the AIM and the RF Local Display Panel.
- **Fallback:** PLC-5 back in service; previous `config.ab` and `srf1.db`.

10.2.4 S3: Heater — Genesys AC under the Legacy Enable

- **Entry:** test-stand acceptance; sync circuit; SW3; a long down.
- **Work:** variac, motor and relay stage replaced; legacy enable and monitors kept.
- **Verification:** warm-up ramp; filament voltage and current at the operating point, inside the limits of Table 7; klystron behavior at turn-on; RF spectrum near the carrier compared with the variac baseline.
- **Fallback:** variac chassis.

10.2.5 S4: Interface Chassis and Waveform Buffer; ARC4 Trip Enabled

- **Entry:** production unit, with both boards and the features for running beside the legacy chain; SW7; RF taps and HVPS signal cables for the Waveform Buffer; O5 and O6 closed, and O3 if SCR ENABLE is regenerated; ARC4 monitor-only record; failure-mode analysis of the chosen insertion.
- **Work:** the chassis installed; Interface Chassis output in series at the chosen point, with jumpers for the absent inputs; ARC4 trip routed through it; Waveform Buffer recording beside the legacy diagnostics, its comparators observed first and enabled after its collector power has been compared with the PLC-5's over a run period.
- **Verification:** every input fault injected; first-fault record checked; legacy trip behavior unchanged; Waveform Buffer channels checked against the legacy diagnostics.
- **Fallback:** bypass jumper; series contact removed.

10.2.6 S5: HVPS Controller and PPS Interface Box

- **Entry:** ADSO approval and final design of the PPS box; SW1 and SW6; legacy database variants or selector; FCOG1200 and regulator, or the decoupled option; O10 closed.
- **Work:** the off-duty unit receives the new controller and its PPS Interface Box in one down and is operated for a run period; the second unit follows at a later down.
- **Verification:** regulation and ripple against the legacy unit; every HVPS interlock and crowbar source; the fiber trio to the RF Local Display Panel; PPS certification; EMERGENCY OFF test.
- **Fallback:** switch to the other unit.

10.2.7 S6: LLRF9 Drives the Klystron

- **Entry:** S4 complete with the LLRF9 enable wired; SW9, with O2 closed; interlocked RF transfer switch; LLRF9 acceptance criteria; O8 closed.
- **Work:** the LLRF9 drives the klystron; the RFP stays as hot standby with the IQAs measuring; the RFP-specific SNL functions are disabled; the coordinator runs turn-on, the HVPS loop and the fault response.
- **Verification:** every RF-off path tested with the LLRF9 driving; stability, ripple rejection and synchrotron-line damping against the legacy baseline [R4], [R12].
- **Fallback:** transfer switch to the RFP; legacy SNL configuration restored.

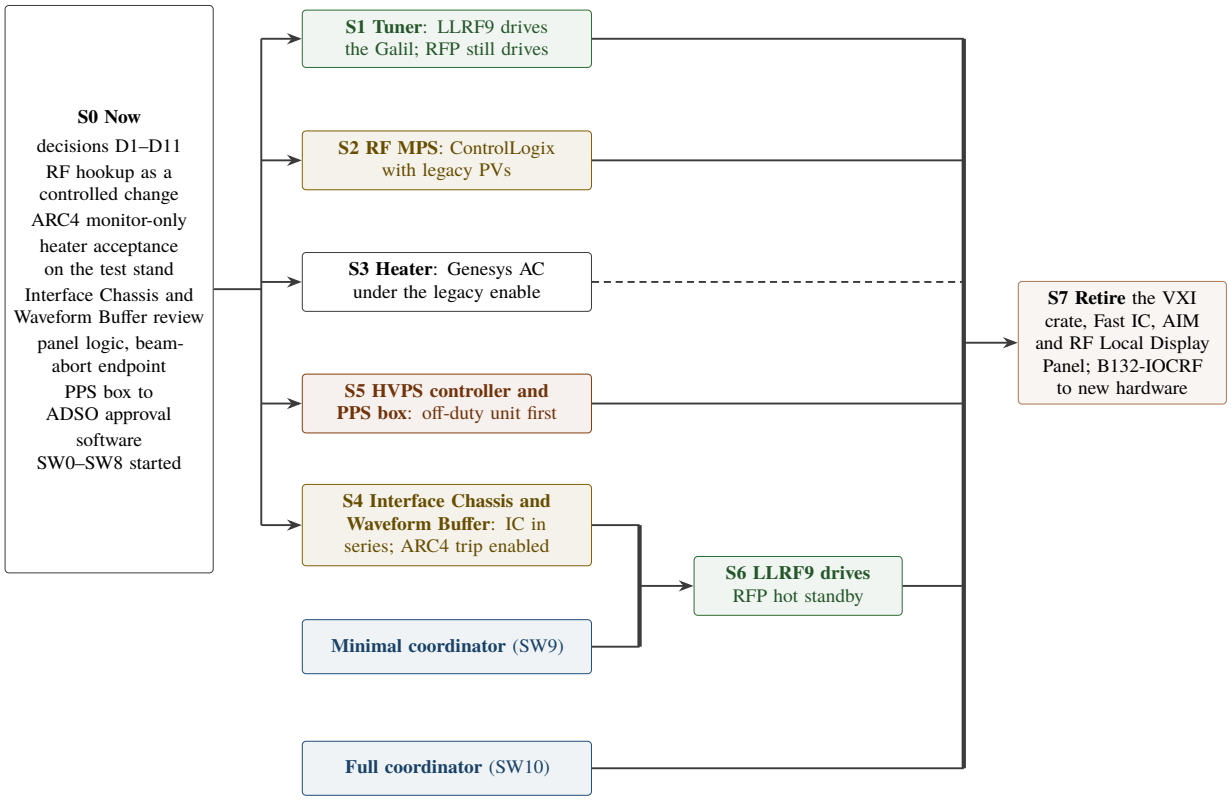


Figure 2: Stage dependencies. Stage S0 feeds the five independent stages S1–S5, which can be ordered freely. A thick vertical bar collects inputs that are all required: S6 needs the Interface Chassis in series (S4) and the minimal coordinator; S7 needs S1, S2, S5, S6 and the full coordinator. The dashed line marks a soft prerequisite (the heater for S7). Not drawn: the soft prerequisite S1 for S6. The two coordinator releases are software streams that start in S0.

10.2.8 S7: VXI Retirement

- **Entry:** S1, S2, S5 (or an RIO bridge) and S6 complete; SW10 tested on the new hardware for B132-IOCRF; every orphan function re-homed, including the beam-abort output (O4); separate reference distribution; Interface Chassis in hub mode with the RF MPS in upgrade mode; a full run period of S6 without a fallback to the RFP.
- **Work:** VXI crate, Fast Interlock Chassis, AIM and RF Local Display Panel removed in one window; B132-IOCRF moved to its new hardware; Interface Chassis takes the hub role; RIO link retired.
- **Verification:** end-to-end interlock test with fault injection on every layer, including the beam abort to the SPEAR MPS; full-power commissioning; operator training.
- **Fallback:** none practical once the hardware is removed, hence the run period required at entry.

10.3 Stage Dependencies

Stages S1 to S5 are independent of each other and can be ordered by installation window and by who is available (Figure 2). S6 needs S4 and the minimal coordinator; S1 is a soft prerequisite, because it gives operating experience with the LLRF9. S7 needs S1, S2, S5 (or an RIO bridge to the legacy SLC-500), S6 and the full coordinator; S3 is a soft prerequisite.

10.4 Installation Windows and People

Windows. Scheduled downs, in particular the April and August downs in which the HVPS units exchange roles [R1]; the summer down for tunnel access [R13]; maintenance days for monitor-only work.

People. The April 2026 schedule names the subsystem leads [R13]; the controls group leads the software work packages [R38]. Their overlaps set the following constraints:

- Controls group (all software work packages, Table 12): one team for every IOC and the coordinator, so the packages follow the stage order; SW5 before SW9.
- M. Larrus (both PLC programs, Test Stand 18): S2 and S5 not in the same window.
- J. Sebek (Interface Chassis, LLRF9, integration tests): Interface Chassis review, panel logic and RF hookup before S6 work.
- B. Morris (PPS box, heater, sync circuit, Test Stand 18): PPS approval before S5; S3 when the sync circuit is ready.
- M. Dunning (motion control, LLRF9 software): S1.
- D. Zhang (Waveform Buffer): production review and S4, with J. Sebek.
- R. Lion and K. Vedyanina (arc detection): ARC4 installation in S0.

The controls group has not yet named an owner for changes to the legacy IOC (SW1).

11 Deferral and Contingency Scenarios

11.1 If a Subsystem Is Deferred

Table 14: Effect of deferring a subsystem.

If this slips	Still possible	Blocked	Mitigation
Interface Chassis production	S1, S2, S3, S5; ARC4 trip at an interim point	S4, with the Waveform Buffer; S6, S7	Interim LLRF9 permit relay box, as an emergency measure only
EPICS-side software	The ARC4 monitor-only installation	Every cutover	Generated databases, standard drivers and vendor software, minimal coordinator first
HVPS firing board or regulator	All other stages	S5, with the PPS box, unless decoupled; S7	CompactLogix first with the existing firing board and regulator; or an RIO scanner module in the RF MPS chassis bridging rack 1
PPS approval	All other stages	S5, S7	Complete the documentation early; the new HVPS PLC must not carry the PPS chains, so S5 waits
RF MPS	S1, S3, S4 (with the legacy AB summary and heartbeat), S5, S6	S7	None: the PLC-5 is the first RIO node and the slow data system
Galil	S6 with the legacy tuner loop on IQA phases	S7	None

If this slips	Still possible	Blocked	Mitigation
Heater	All	None, if the RF MPS keeps the legacy heater I/O	None
ARC4	All	None	Jumper the Interface Chassis input
LLRF9	S2 to S5	S1, S6, S7	The Interface Chassis and Waveform Buffer still add protection and diagnostics
Minimal coordinator	S1 to S5	S6, S7	Expert-run LLRF9 tests, as in 2021

11.2 If Legacy Hardware Fails During the Transition

Table 15: Contingency paths for legacy failures.

Failure	Effect	Contingency path
VXI CPU	Station down, including the coordinator	A spare CPU; moving B132-IOCRF to its new hardware early needs the S7 prerequisites
VXI module (scanner, CLK, RFP, IQA, AIM)	Station down	LLRF9 drive configuration (S6 prerequisites), with an interim permit adapter if the Interface Chassis is not yet installed
RF Local Display Panel	Station down	Interface Chassis taking over SCR ENABLE, KLYSTRON CROWBAR and the beam abort (needs O3 and O4)
PLC-5	Station down	ControlLogix with the legacy-faithful program and IOC (S2 prerequisites)
SLC-500 of the active HVPS unit	Unit down	Other unit; upgraded unit once the S5 prerequisites exist
Tuner controller or driver	Tuning lost	LLRF9 driving the Galil (S1 prerequisites)
Variac or its motor	Heater control lost	Genesys AC (S3 prerequisites)

Preparing each stage's prerequisites therefore also builds an emergency path for the legacy element it replaces, which is one more reason to keep the new subsystems compatible with the legacy station.

12 Risks

Table 16: Integration and migration risks.

#	Risk	Consequence	Mitigation
1	No owner or build environment for the legacy IOC	Every Class B cutover and the minimal coordinator blocked	Name an owner; set up a test IOC; prefer database-only changes

#	Risk	Consequence	Mitigation
2	Software capacity	Hardware waits for software	Generated databases, standard drivers and vendor software, two coordinator releases
3	Hidden behavior of the PLC-5 program	Unexpected trips or a missing interlock after S2	Legacy-faithful mode, legacy data image, full trip matrix
4	RF hookup changes legacy calibrations	Protection thresholds move	Controlled change with recheck of thresholds
5	Drive-source mix-up	Klystron driven by an unintended source	Interlocked transfer switch with readback
6	Loss of fast RF cut with the LLRF9 driving	Delayed RF removal after a SPEAR MPS or orbit trip	Interface Chassis and minimal coordinator as hard prerequisites of S6
7	PV name collisions	Undefined Channel Access behavior	Naming policy (SW0)
8	RIO chain broken during a cutover	Loss of HVPS and tuner data	Splice prepared and tested; keep rack terminals connected
9	PPS approval delayed	HVPS controller blocked, because the two go in together	Complete the documentation and submit it for ADSO approval early
10	Same people needed for several stages	Stages slip together	Window plan by person (§10.4)
11	Nuisance trips from the ARC4	Lost beam time; trip disabled	Monitor-only record before enabling
12	A solid-state amplifier project starts during the migration	Priorities and people diverted	Keep the stages independent; the Interface Chassis and coordinator design do not depend on the RF source [R26]
13	Orphan functions discovered late	VXI retirement delayed	Table 11 reviewed with operations and the SPEAR MPS group
14	Tuner control depends on the LLRF9 IOC	No tuning while it is down	Legacy tuner hardware kept connected as fallback; spare LLRF9 units [R42]; IOC health alarm
15	RF Local Display Panel logic not recovered	Interface Chassis behaves differently from the legacy chain	Read the device or test the truth table early (O3)

13 Decisions Needed

Table 17: Decisions needed.

ID	Decision	Recommendation	Needed by
D1	Migration strategy	Incremental cutovers compatible with the legacy station, each with a fallback (§10)	Now
D2	PV naming policy	Legacy names stay with the legacy IOC; new IOCs use non-colliding prefixes; recorded exceptions (§9.6)	Before SW5
D3	Tuner control	Decided: the LLRF9 tuner loops drive the Galil directly; no separate Galil IOC [R38]	Decided

ID	Decision	Recommendation	Needed by
D4	RF MPS EPICS path	Re-pointed legacy records with a new IOC; RIO emulation as an alternative	Before SW5
D5	Legacy data image	In both new PLC programs	Before SW5 and SW6
D6	Interface Chassis production scope	Include the features for running beside the legacy chain and the beam-abort output (required [R38]); keep the RF Local Display Panel during the transition	Production review
D7	ARC4 interim trip point	From the Fast Interlock Chassis and AIM drawings	Before the ARC4 trip is enabled
D8	Heater interim control	Legacy enable and monitors; ramp by soft start or IOC	Before S3
D9	HVPS staging	Whether to install the CompactLogix first with the existing firing board and regulator	Before S5
D10	Coordinator releases	Minimal (S6) and full (S7), in the IOC B132-IOCRF, with per-subsystem adapters	Now
D11	Legacy IOC owner	Name one in the controls group	Now

14 Actions

Table 18: Actions.

#	Action	Suggested lead	For
A1	Record decisions D1–D11	Project lead	S0
A2	Name a legacy IOC owner; set up a test IOC; check the SNL build	Controls group	SW1
A3	Plan the RF hookup as a controlled change, including the threshold recheck	J. Sebek	S0
A4	Install the ARC4 monitor-only	R. Lion	S0
A5	Hold the production review of the Interface Chassis and Waveform Buffer with the checklist of §6.3, and add the beam-abort output to the specification	J. Sebek, D. Zhang	S4
A6	Recover the RF Local Display Panel logic (O3)	J. Sebek	S4, S7
A7	Trace the beam-abort output to the SPEAR MPS and agree its signal convention (O4)	Project lead with the SPEAR MPS group	S7
A8	Complete the PPS Interface Box documentation and final design; obtain ADSO approval	B. Morris	S5
A9	Complete the heater test-stand acceptance and the sync circuit	B. Morris	S3
A10	Measure the legacy heartbeat; trace the Slot 5 permit loop	J. Sebek	S4
A11	Run the PLC-5 trip matrix	S. Condamoor	S2
A12	Recover the 2021 LLRF9 configuration	J. Sebek	S6

#	Action	Suggested lead	For
A13	Configure the LLRF9 tuner loops and Galil driver; bench-test with the DMC-4143; run the shadow comparison	M. Dunning	S1
A14	Write the HVPS unit transfer list and the switch-over procedure	M. Larrus	S5
A15	Ask Rockwell about RIO adapter emulation	S. Condamoor	D4

15 Remaining Recommendations for Doc 0

Doc 0 R4 [R1] follows the PV naming policy of §9.6. Two recommendations remain for its next revision:

1. Add the orphan functions of Table 11 to the allocation of functions.
2. Add a transition section that refers to this plan.

A Source Index

Ref	Source	Location
[R1]	Doc 0, SPEAR3 LLRF Upgrade System Design Report, R4	`Designs/tex/doc0-system-design/`
[R2]	Doc L, Legacy System Architecture, R1	`Designs/tex/docL-legacy-architecture/`
[R3]	Doc I, Legacy Interlock Architecture, R0	`Designs/tex/docI-interlock-architecture/`
[R4]	Doc P, RF Physics and Plant, R0	`Designs/tex/docP-rf-physics/`
[R5]	Doc T, Tuner Control System Analysis, R0	`Designs/tex/docT-tuner-control/`
[R6]	J. Sebek, LLRF Interface Chassis specification	`llrf/architecture/llrfInterfaceChassis.docx`
[R7]	J. Sebek, fiber-optic connections between the RF system controllers (HVPS controller side)	`hvps/architecture/designNotes/controllerFiberOpticConnections.docx`
[R8]	J. Sebek, RF system MPS requirements	`hvps/architecture/designNotes/RFSystemMPSRequirements.docx`
[R9]	J. Sebek, LLRF documentation notes, R2 (RIO topology, fibers, RF Local Display Panel, Fast Interlock Chassis)	`llrf/documentation/LLRFDocumentationNotesR2.docx`
[R10]	RF Local Display Panel to cross-connect mapping (SD-340-311-01 against WD-340-330-02)	`llrf/documentation/LocalPanelToXConnectMapping.xlsx`
[R11]	B132-11 patch panel configuration and coupler losses	`llrf/calibrations/b132R11PatchPanel.xlsx`
[R12]	J. Sebek, comparison of the LLRF9 with the legacy LLRF system (2021)	`llrf/tests/llrf9Tests.tex`
[R13]	SPEAR3 LLRF Upgrade project schedule, April 2026	`Designs/docx/SPEAR3_LLRF_Upgrade_Project_Schedule.xlsx`
[R14]	Legacy IOC startup script, head revision 1.18	`spear-rf-code-legacy/iocBoot/`
[R15]	Legacy RIO scanner configuration (`config.ab`,v)	`spear-rf-code-legacy/iocBoot/`
[R16]	Legacy HVPS database (`rf_hvps.db`,v)	`spear-rf-code-legacy/rfApp/`

Ref	Source	Location
[R17]	Legacy analog channel template and substitutions (`rf_analog.db`, `rf_analog_All`, `rf_analog_4CV`)	`spear-rf-code-legacy/rfApp/`
[R18]	Legacy temperature, interlock and digital substitutions	`spear-rf-code-legacy/rfApp/`
[R19]	Legacy tuner loop PV definitions (`rf_tuner_loop_pvs.h,v`)	`spear-rf-code-legacy/rfApp/`
[R20]	Legacy HVPS loop PV definitions (`rf_hvps_loop_pvs.h,v`)	`spear-rf-code-legacy/rfApp/`
[R21]	Legacy station state machine (`rf_states.st,v`)	`spear-rf-code-legacy/rfApp/`
[R22]	Legacy cavity database with the tuner records (`rf_cav.db,v`, `rf_stn_cav.db,v`)	`spear-rf-code-legacy/rfApp/`
[R23]	Legacy cavity and station summaries (`rf_sumy_cav.db,v`, `rf_sumy_stn.db,v`)	`spear-rf-code-legacy/rfApp/`
[R24]	LLRF9 tuner loop and Galil screens (`tuner.edl`, `motor_galil.edl`)	`llrf/llrf9/iGp/dl_llrf/`
[R25]	Klystron filament heater control chassis drawing	`llrf/documentation/filamentHeater/sd3403110002.pdf`
[R26]	SSA upgrade summary, v11 (Test Stand 18 status, proposed SSA project)	`SSA_upgrade/`
[R27]	J. Sebek, interfaces between the RF system controllers	`hvps/architecture/designNotes/interfacesBetweenRFSystemControllers.docx`
[R28]	J. Sebek, SPEAR3 RF station operation	`llrf/documentation/LLRFOperation_jims.docx`
[R29]	Hoffman box PPS wiring	`pps/HoffmanBoxPPSWiring.docx`
[R30]	RF Local Display Panel drawings: drawing list DL-340-310-00, front and rear panels SI-340-310-05 and -02, schematic SD-340-311-01 (“PEP-II RF System Local Panel”)	`llrf/documentation/localPanel/`
[R31]	Fast Interlock Chassis schematics SD-340-308-01 and -02	`llrf/documentation/legacyInterfaceModules/`
[R32]	AIM schematic SD-340-309-01	`llrf/documentation/legacyInterfaceModules/sd3403090102.pdf`
[R33]	Wiring diagram WD-340-330-29 (control fiber-optic cables of the RF Local Display Panel)	`llrf/documentation/mpsWiringDiagrams/wd3403302900.pdf`
[R34]	Wiring diagram WD-340-330-02 (RF Local Display Panel cross-connects)	`llrf/documentation/mpsWiringDiagrams/wd3403300200.pdf`
[R35]	HVPS switch-over procedure, R0	`hvps/documentation/procedures/spearRfHvpsSwitchProcedureR0.docx`
[R36]	Legacy RIO adapter substitutions (`rf_ab_4CV.substitutions,v`)	`spear-rf-code-legacy/rfApp/`
[R37]	Galil test records (2024–2025) and commissioning notes	`llrf/tuners/galil/`
[R38]	Project status and decisions reported by the project lead, 31 October to 3 November 2026	Project records

Ref	Source	Location
[R39]	J. Sebek, fiber-optic connections between the LLRF system and the HVPS controller, Rev 3	`llrf/documentation/fiberOpticCableSignalControlRev3.docx`
[R40]	Dimtel, LLRF9 manual (SPEAR3 screens)	`llrf/llrf9/llrf9_manual_s3_screen.pdf`
[R41]	Legacy tuner loop program (`rf_tuner_loop.st`,`v`)	`spear-rf-code-legacy/rfApp/`
[R42]	J. Sebek, LLRF upgrade task list, Rev 3	`llrf/documentation/LLRFUpgradeTaskListRev3.docx`

B Legacy EPICS Interface Details

All record names carry the prefix `SRF1:`.

B.1 HVPS Records on RIO Rack 1

Table 20: HVPS records on RIO rack 1 [R16].

Record	Type	Function	Address
`HVPS:CURR`	ai	Measured current	A1 C8
`HVPS:VOLT`	ai	Measured voltage	A1 C6
`HVPS:VOLT:RBCK`	ai	Voltage setpoint readback	A1 C4
`HVPS:VOLT:ULIM`	ai	Voltage limit readback	A1 C10
`HVPSAC:CURR`	ai	AC line current	A1 C2
`HVPS:VOLT:CTRL`	ao	Voltage setpoint, written by the HVPS loop	A1 C2
`HVPS:VOLT:MAX`	ao	Voltage limit	A1 C4
`HVPS:RESET:CTRL`	bo	Reset	A1 C10
`HVPSCONTACT:CLOSE:CTRL`	bo	Contactor close	A1 C6
`HVPSSCR:ON:CTRL`	bo	SCR enable, set by the state machine	A1 C8
`HVPSDCM:T0:STAT`	mbbi	DCM table status	A1 C0

In addition, 32 status bits of the HVPS controller are read from rack 1 through the digital templates [R18], and the HVPS oil temperature comes from the PLC-5.

B.2 Template Instances Fed by the PLC-5

Table 21: Legacy template instances fed by the PLC-5 [R17], [R18].

Template	Content	Instances
`rf_temp`	Temperatures	111
`rf_interlock`	Interlock bits	41

Template	Content	Instances
`rf_analog`	Analog channels with PLC-5 trip-limit readbacks, including tuner positions, filament, focus, tuner supply and the HVPS voltage and current seen by the PLC	20
`rf_analog_log`	Logarithmic analog channels (vacuum, ion pumps)	8
`rf_sumy_plc`	PLC summaries	24
`rf_ab_module`	PLC module status	16
`rf_digital_plc`	PLC digital status	12
Total		232

Counts are from the head revisions of the substitution files loaded by the IOC. Each instance holds several records (value, alarm latch, summary links). A few further records in the klystron, station and HVPS databases also read the PLC-5.

B.3 SNL Bindings Relevant to the Migration

Table 22: SNL bindings relevant to the migration.

Program	PVs	Use	Effect of the migration
HVPS loop	`STN:RFP:MODU.DLE`, `STN:RFP:MODU.SEVR`	Regulation mode; RFP health	With the RFP idle the loop regulates on the wrong variable or alarms [R20]
HVPS loop	`HVPS:VOLT:CTRL`, `HVPS:VOLT`	Setpoint and readback	Re-pointed to the HVPS IOC [R20]
HVPS loop	`KLYSOUTFRWD:POWER`, `KLYSDRIVFRWD:POWER:ERR`, `STN:VOLT`	IQA inputs	Must stay valid while the legacy HVPS loop runs [R20]
Tuner loop	`CAVnTUNR:STEP:MOTOR` fields `RBV`, `DRVH`, `DRVL`, `DMOV`, `RDBD`	Motor readback, limits, done flag, deadband	Unused with the LLRF9 driving the Galil; kept for the fallback [R19], [R22]
Tuner loop	`CAVnTUNR:POSN:CTRL`	Target position	Unused with the legacy loops off [R19]
Tuner loop	`CAVnTUNR:POSN`	Potentiometer readback from the PLC-5	Kept through the RF MPS cutover [R19]
Tuner loop	`KLYSOUTFRWD:POWER:MIN`, `CAVnLOAD:ANGLE:ERR`	Minimum power; phase error	From the IQAs [R19]
State machine	`CAVTUNR:LOOPON:RESET`, `CAVTUNR:LOOPPARK:RESET`	Tuner home and park, then a fixed wait	No effect on the tuners driven by the LLRF9 [R21]
State machine	`STN:AIM:FRCBMABT`, `STN:AIM:MODU.RBA`	Force and reset of the beam abort	Replaced by the coordinator's force and reset of the Interface Chassis output [R21]

C RF Local Display Panel I/O

The RF Local Display Panel 340-310 sits in rack B132-12 and connects to cross-connect block X530 in rack B132-14 through two 25-pin connectors [R9], [R10], [R30], [R34]. Its eleven fibers are listed in Table 23 [R9], [R30], [R33].

Table 23: Fibers of the RF Local Display Panel.

Panel device	Panel label	Light means	Other end
U10	VXI beam abort	No abort	Fast Interlock Chassis (BEAM_ABORT)
U11	HVPS beam abort	No abort	HVPS controller (STATUS)
U12	HVPS on request	Request	Fast Interlock Chassis (HVPS_ON)
U13	Station online request	Request	Fast Interlock Chassis (FILAMENT_ON)
U14	Filament and focus request	Request	Fast Interlock Chassis (SOLENOID_ON)
U15	Filament timeout bypass	Bypass	Fast Interlock Chassis (FILAMENT_TIMEOUT)
U16	Reset	Reset	Fast Interlock Chassis (FAULT_RESET)
U19	HVPS triggers enable	HVPS on	HVPS controller (SCR ENABLE)
U20	HVPS crowbar	No crowbar	HVPS controller (KLYSTRON CROWBAR)
U21	AB heartbeat	Pulsing	Fast Interlock Chassis (HEARTBEAT)
U22	AB summary	No fault	Fast Interlock Chassis (A/B SUMMARY)

The labels below are those of the panel schematic SD-340-311-01 with the cross-connect labels of WD-340-330-02; the source does not state signal directions. The labels suggest that the “Req” lines carry requests from the panel to the PLC-5 and the “Ctrl” lines carry the PLC-5’s commands.

Table 24: RF Local Display Panel connector J2.

Pins	SD-340-311-01 label	WD-340-330-02 label
J2-01, J2-14	Vin+, Vin− (from HVPS)	HVPS Vin+, Vin−
J2-02, J2-15	Iin+, Iin− (from HVPS)	HVPS Iin+, Iin−
J2-03, J2-04	Vout, Iout (to AB)	HVPS Vout, Iout
J2-06	SolenoidM Ctrl24 (AB)	Main Focus On
J2-07	SolenoidM Fault15 (supply)	Main Focus PS Interlock
J2-09	Filament Ctrl24 (AB)	Filament On
J2-10	Timeout Ctrl24 (AB)	Filament Timer Bypass

Pins	SD-340-311-01 label	WD-340-330-02 label
J2-11	HV Detected24 (AB)	HV Detected
J2-12	Local Mode24 (AB)	Local/Remote
J2-18	Solenoid Req24	Online/Offline Request
J2-19	SolenoidB Ctrl24	Bucking Coil PS On/Off
J2-20	SolenoidB Fault15 (supply)	Bucking Coil PS Interlock
J2-21	Filament Req24	Filament PS On/Off
J2-22	Timeout Req24	Filament Timer Bypass
J2-23	Reset Req24	Interlock Reset
J2-24	RF detected24	RF On Indicator
J2-25	Online mode24	Online/Offline Control
J2-05, -08, -13, -16, -17	Returns	Returns

Connector J3 carries 24 V summary lines from the PLC-5 (SD-340-311-01 pins): cavity 1 to 4 (J3-01, J3-14, J3-02, J3-15), klystron (J3-03), focus (J3-04), magic tee 1 to 3 (J3-17, J3-05, J3-18), waveguide pressure (J3-06), water flow (J3-07), temperature (J3-08), collector (J3-16), circulator (J3-19) and vacuum (J3-20); the AB summary (J3-09) and the AB heartbeat (J3-21); spares on J3-10, J3-11 and J3-22 to J3-25; ground on J3-12 and J3-13. The cross-connect diagram numbers the same signals in reverse pin order (for example, its J3-05 “Interlock Sum” corresponds to J3-09 and its J3-18 “AB Heartbeat” to J3-21), and it assigns “Waveguide Air MPS” to the line that the schematic shows as spare 1 (J3-22); J. Sebek records the same discrepancy [R39]. Any connection at J3 is checked on the hardware first.

D Abbreviations

Abbreviation	Meaning
AB	Allen-Bradley
ADSO	Accelerator Directorate Safety Office (SLAC)
AIM	Arc Interlock Module (VXI slot 12)
CLK	VXI clock and RF reference distribution module
DCM	Direct Communication Module (1771-DCM, 1747-DCM)
EPICS	Experimental Physics and Industrial Control System
Fast IC	Fast Interlock Chassis 340-308
GAL	Generic array logic, a programmable logic device
HV, HVPS	High voltage, high-voltage power supply
IOC	EPICS Input/Output Controller
IQA	In-phase, quadrature and amplitude detector module
LLRF	Low-level RF
MPS	Machine protection system
PPS	Personnel protection system

Abbreviation	Meaning
PV	EPICS process variable
RFP	RF Processor module
RIO	Allen-Bradley Remote I/O
SNL	State Notation Language (EPICS sequencer)
SSA	Solid-state amplifier
TID	Technology and Innovation Directorate
VXI	VME eXtensions for Instrumentation